

THE ENERGY FOOTPRINT OF AI & DATA CENTERS

Scale, Cost, and Growth of Computing Power Consumption



Annual Electricity Consumption



Data Centers 2024

415 TWh

Data Centers 2030 ↗

945 TWh (projected — capacity nearly doubles)



France

430 TWh



United Kingdom

300 TWh



Sweden

140 TWh



Argentina

130 TWh



Netherlands

120 TWh

Data Centers ≈ France's entire grid

1.5% of global electricity now → ~3% by 2030



Training ONE Large AI Model

GPT-3: True environmental cost of a single training run



1,287 MWh

Electricity consumed in one training run

Equivalent to:



120 years

US household electricity



112 cars

driven 1 year (CO₂ equiv.)



552 metric tons CO₂ equivalent

Energy Per Query

AI vs. Traditional Web Search

Google Search

0.04 Wh

7.5x

more energy

ChatGPT (GPT-4)

0.30 Wh

Annual ChatGPT: ~227 GWh = 20,000 US homes powered for a year



AI-Driven Growth: Where Are We Headed?

2024

415 TWh
1.5% of global

AI servers: 30% / year
Conventional: 9% / year
Capacity doubling in 6 years

2030

945 TWh
~3% of global



AI server demand grows 30x faster than the global grid



= Japan's entire electricity grid